Project Proposal

Implementing Imaginary time - TDDFT for periodic systems

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Proposal

My project is based on the paper 'Imaginary-time time-dependent density functional theory for periodic systems' by McFarland and Manousakis (2021). The authors devised this method to address limitations in standard Density Functional Theory (DFT) solvers. While the traditional Self-Consistent Field (SCF) iterative method is the standard for finding ground states, it frequently encounters convergence difficulties in specific systems—such as charge sloshing in large metallic systems—or algorithmically converges to an excited state rather than the true ground state[1] . Imaginary-Time TDDFT (it-TDDFT) resolves this by propagating the Kohn-Sham orbitals in imaginary time, a process which naturally decays high-energy components and forces the system toward the true ground state. This was implemented in fortran which is the native language of Quantum ESPRESSO, I will be implementing it in python on Quantum MAterialS Ab initio eLectronic-structure pAckage (Quantum MASALA). This package contains tools for density functional theory (within the Kohn Sham framework) for solids, time-dependent density functional theory and many body perturbation theory.

Theoretical Background

The standard Self-Consistent Field (SCF) method searches for the ground state through an iterative process: guessing an initial charge density, constructing the Kohn-Sham Hamiltonian, diagonalizing it to find the eigenvectors, and using those to calculate a new density . The imaginary-time time-dependent density functional theory (it-TDDFT) approach bypasses this computationally expensive diagonalization step by utilizing imaginary-time evolution. Ref. 1

By substituting real time with imaginary time ($\tau = it$), the standard oscillatory time-evolution is mathematically transformed into an exponential decay process. The evolution of the Kohn-Sham orbitals is governed by the differential equation :

$$\frac{\partial}{\partial \tau}|\psi_m(\tau)\rangle = -H[\rho(\tau)]|\psi_m(\tau)\rangle$$

The physical mechanism driving this convergence becomes clear when an arbitrary initial guess state is expanded as a linear combination of the exact Hamiltonian eigenstates, $|\phi_n\rangle$, with corre-

[1] C. Flamant, G. Kolesov, E. Manousakis, and E. Kaxiras, "Imaginary-Time Time-Dependent Density Functional Theory and Its Application for Robust Convergence of Electronic States," *J. Chem. Theory Comput.* **15**, 6036–6045 (2019). https://doi.org/10.1021/acs.jctc.9b00617

sponding energies E_n. Under continuous imaginary-time propagation, the state evolves as:

$$|\psi(\tau)\rangle = \sum_n c_n e^{-E_n \tau} |\phi_n\rangle$$

Because $E_n > E_0$ for all excited states, the higher-energy components decay exponentially faster than the ground state. Provided the initial guess wavefunction has a non-zero overlap with the true Kohn-Sham ground state, the wavefunction naturally purifies and converges strictly to the exact ground state.

Furthermore, unlike standard quantum mechanical time evolution governed by a unitary operator (e^{-iHt}), imaginary-time propagation is inherently non-unitary. This non-unitary evolution does not preserve the norm or the orthogonality of the propagated states . Left unconstrained, all orbitals would mathematically collapse into the single lowest-energy state, violating the Pauli exclusion principle. Consequently, it is a strict theoretical requirement to explicitly re-orthonormalize the orbital subspace during propagation to maintain a physically valid fermionic system.

Numerical techniques

The numerical techniques that will be implemented in this project will be

- Forward Euler method:
  I will be using this method to find the wavefunction at time $\tau + \Delta\tau$

  $$|\psi_m(\tau + \Delta\tau) = (\mathrm{I} - \Delta\tau \mathrm{H}[\rho(\tau)])|\psi_m(\tau)\rangle$$

- Gram Schmidt Orthogonalization

  $$|\psi_m\rangle \Rightarrow |\psi_m\rangle - \sum_{n<m} |\psi_n\rangle\langle\psi_n|\psi_m\rangle,$$

  $$|\psi_m\rangle \Rightarrow \frac{|\psi_m\rangle}{\sqrt{\langle\psi_m|\psi_m\rangle}}.$$

- Stability Analysis
  To prevent the propogation from diverging exponentially the imaginary time-step $\Delta\tau$ must be constrained. It is set dynamically or static based on the maximum kinetic energy cut off of the plane wave basis.

Validation and Benchmarking

First, the algorithm will be implemented for a 1D Quantum Harmonic Oscillator, a system with analytically known ground and excited states. This ensures the underlying mathematical logic and normalization routines are sound. Following this, the method will be integrated into Quantum MASALA to simulate a 3D diamond crystal. Because the original authors successfully benchmarked diamond, silicon and graphene, their published energy and charge-density convergence plots will serve as a direct reference to validate the accuracy of this implementation[2].

[2] K. P. McFarland and E. Manousakis, Phys. Rev. B 104, 245116 (2021).